
ITECH 2001 PART B: PROTOTYPE PLAN

DESERT ODYSSEY

Subject Name: Game Development Fundamentals

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Generate Scenario

Your task this semester is to pitch, plan and develop a single scene for a game prototype with an **Exploration** theme.

It must be set in a 3D world, in a **Desert** environment, with one appropriate **2D GIMP texture that can be applied to your environment** created by yourself.

It must be set during **Night** time.

You are required to create two **Blender** 3D objects yourself, a **Park Bench** and an **additional static 3D object of your own choice** that suits your game scene.

The player/camera viewpoint should use a **Third Person camera**.

The player can approach another character and trigger an event with **Friendly** dialogue.

There should be additional events: a **Player Respawn** and an **Event of your own choice** (*must be different to your other events*). When triggered, each event should update the interface, such as displaying information, score, lives, etc.

High Concept Statement

"Desert Odyssey" is a 3D exploration game set in a vast desert at night, where players guide Amir through a series of challenges to reach a distant watchtower. The game emphasizes exploration, interaction with NPC, and overcoming environmental hazards like a sandstorm, with the ultimate goal of reaching the watchtower for safety.

Mechanics

Space

The prototype level unfolds in a 3D desert environment at night, navigated through a third-person perspective. Players move by walking or sprinting across sand dunes, rocky outcrops, and sparse desert vegetation. The level's boundaries are defined by mountains to the north, a watchtower to the east, and scattered dunes elsewhere. Key elements include interactive objects like a park bench and Zara, a friendly NPC. Dynamic events, such as a sandstorm, challenge navigation. The main objective is to guide Amir to the watchtower, engaging in dialogue and overcoming obstacles. The design draws inspiration from exploration games like "Journey."



Figure 1: Journey (Thatgamecompany, 2012)

Time

The game unfolds entirely at night, amplifying the sense of mystery and adventure in the desert environment. The limited visibility requires careful navigation, with moonlight and subtle glows from objects guiding the player. Dynamic events, such as a sandstorm, are triggered at specific intervals, adding a time-based challenge. Time progression influences the player's interaction with Zara, the NPC, whose helpful hints may only be accessible during certain moments.

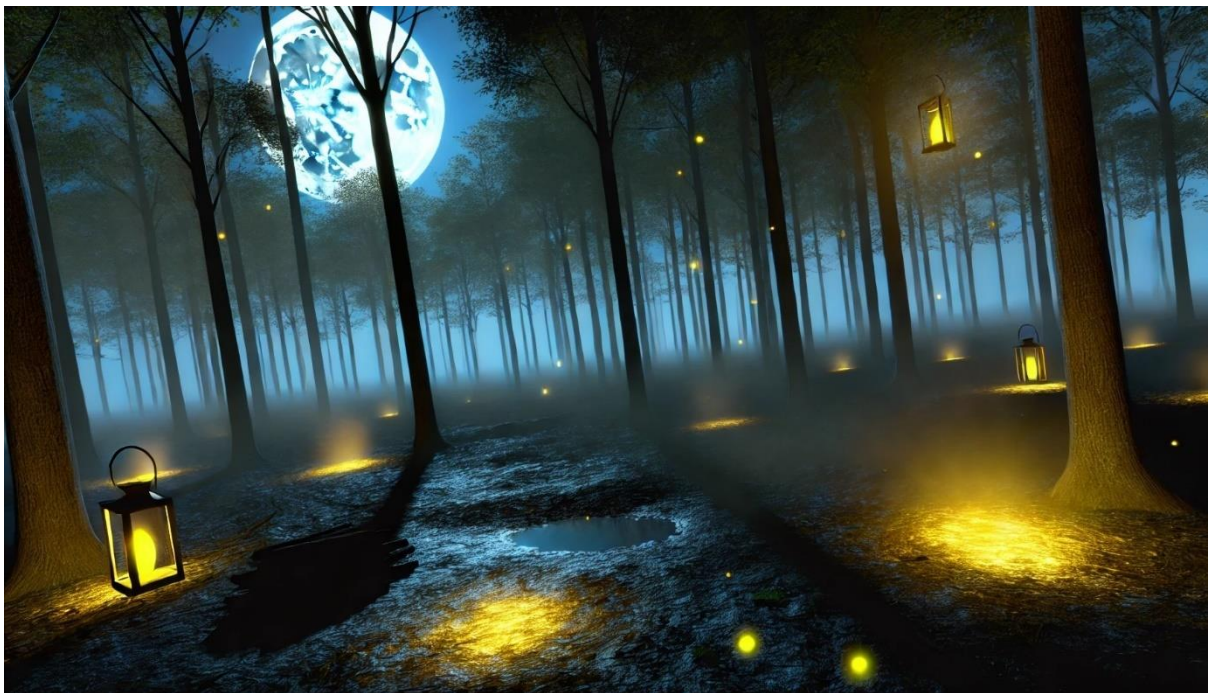


Figure 2: Nighttime settings (DALL-E)

Objects

1. Character:

- **Amir (Player Character)**

Role: Protagonist controlled by the player.

Description: Amir is a modern explorer dressed in practical light desert gear suitable for the harsh desert environment. His attire includes a rugged jacket, utility pants, and sturdy boots. Equipped with minimal tools, Amir is ready to navigate through the treacherous desert landscape.

Purpose: The player controls Amir, guiding him through the expansive desert terrain while interacting with various elements like NPC and overcoming natural and environmental obstacles. Amir's journey is about survival, exploration, and uncovering secrets hidden within the desert

The character's appearance is inspired by Nathan Drake from the video game "Uncharted 3: Drake's Redemption" as shown in Figure 3.



Figure 3: Uncharted 3:

Drake's Redemption (Naughty Dog, 2011)

- **Zara (Non-Playing Character)**

Role: Friendly NPC who provides hints and guidance.

Description: Zara is a welcoming desert spirit embodied in a human-like form, radiating a soft glow, dressed in traditional desert attire. She appears near specific points of interest, such as the park bench, to provide helpful hints and friendly dialogue when approached by Amir. Zara's presence is marked by a warm, inviting aura that signifies a safe interaction point for players.

Purpose: Zara serves to make the game more accessible to beginners by offering timely hints, guidance, and directions, helping players overcome challenges, and ensuring they remain on the right path toward their ultimate goal.



Figure 4: Zara (DALL-E)

2. Props:

- **Park Bench**



Figure 5: Park Bench (Self-made Blender asset)

Description: The park bench is a simple wooden structure, partly buried in the sand and weathered by the desert sun and wind. The planks are cracked, with faded paint and a few missing bolts, showing its age and use over time. Despite its worn-out appearance, the bench stands out in the vast expanse of sand and dunes. It's positioned intentionally near the game's starting point, just beyond the first dunes. The bench offers a familiar and comforting spot in the harsh desert, where Amir can pause and gather his thoughts before moving on. A few scattered rocks and patches of dry grass around the bench suggest that others have rested here before.

Purpose: The park bench plays several important roles in the game. As Amir approaches, he meets Zara, the friendly NPC, and this triggers a dialogue event where she gives valuable hints and directions to help navigate the desert's challenges. The bench serves as a mental anchor for the player—a place to pause, feel safe, and prepare before exploring deeper into the desert. It breaks up the emptiness of the desert, providing a brief moment of rest in the journey. The bench also helps the player feel oriented within the game world, acting as a landmark to plan their next steps and maintain a sense of direction in the vast desert.

- **Power-Up: Water Cache**



Figure 6: Water Cache (DALL-E)

Description: The "Water Cache" is a temporary power-up that players can find scattered throughout the desert. These caches are hidden in small, inconspicuous locations, like beneath a rock, inside a hollow log, or near an old ruin. Represented by a glowing water droplet icon, they stand out against the sandy terrain.

Purpose: The Water Cache provides instant hydration, immediately refilling Amir's hydration meter. This power-up offers a quick boost to sustain Amir during difficult stretches, such as navigating the sandstorm or when far from a regular water source. It adds an additional layer of exploration and reward, motivating players to search every corner of the map for these helpful items.

3. Interface

- **Dialogue Interface**

Description:

The dialogue interface is a user interface element that appears on the screen whenever Amir interacts with another character, such as Zara, the friendly NPC. It displays the dialogue being spoken by Zara along with a set of response options for Amir. These options are usually presented as text choices that the player can select to guide the conversation.

Purpose:

The dialogue interface facilitates interaction between the player and the game world by allowing players to actively participate in conversations with the NPC. It offers multiple response options, enabling players to choose how Amir responds to Zara's hints or guidance. The choices made can influence the direction of the dialogue, reveal additional hints, or provide background information about the game world. The dialogue interface also adds depth to the narrative, making the player's decisions feel meaningful and connected to the overall journey through the desert. It enhances immersion by making players feel more involved in the story and the outcomes of their choices.

- **Sandstorm Event**



Figure 7: Sandstorm event (Fortnite: Battle Royale, 2024)

Description:

The sandstorm is a tough environmental challenge that hits Amir as he approaches the mountain area of the desert. Suddenly, a fierce storm sweeps across the landscape, filling the air with swirling sand and dust that makes it hard to see and find the way forward. The sandstorm causes everything to blur and darken, with fast-moving particles flying across the screen and the sound of howling winds adding to the chaos.

Purpose:

The sandstorm plays several important roles in the game. It creates a time-based challenge that forces the player to quickly think about their next move. With reduced visibility, players need to carefully choose their path, using familiar landmarks like the park bench or the watchtower to stay on track. The storm also adds a sense of urgency and danger—if Amir

stays in the storm too long, he risks losing health or getting lost, which could lead to a respawn at the last safe location.

The event pushes players to explore and think strategically, finding shelter or reaching the next goal as quickly as possible. Zara may offer advice on how to get through the storm, making her guidance even more valuable. Overall, the sandstorm adds tension and excitement, making Amir's journey through the desert feel more intense and unpredictable.

4. Camera

- **Third Person Camera**

The player's viewpoint for controlling Amir

5. End Point

- **The Watch Tower**



Figure 8: The Watch Tower (Self-made Blender Asset)

Description:

The watchtower is a tall, cylindrical stone structure that stands out in the desert landscape and serves as the final destination in "Desert Odyssey." Built from old, weathered stones, the tower shows its age with cracks along its walls and a crumbling top. Narrow windows are placed at different heights, offering views of the desert from inside. The tower's partially intact conical roof adds to its ancient and mysterious look. Positioned at the highest point in the desert, it overlooks the surrounding dunes and mountains and can be seen from almost anywhere in the game world. The watchtower acts as a guiding landmark, symbolizing safety and achievement for Amir.

Purpose:

The watchtower is the final goal of Amir's journey, marking the end of his challenging trek through the desert. Reaching the watchtower means the player has successfully navigated obstacles like sandstorms and managed resources, such as water, along the way. It serves as the climax of the game, where the story may wrap up with a final interaction or reveal. The watchtower also helps players stay oriented, providing a clear point to aim for throughout the game. It could also trigger the final dialogue with Zara or offer a sweeping view of the path Amir has travelled, tying together the exploration, story, and challenges in "Desert Odyssey" and providing a satisfying goal and reward for the player.

Actions

Basic Movement:

Players control Amir's movement using the standard WASD keys on the keyboard. The character can walk or sprint to navigate through the desert landscape and interact with objects.

- **Move Forwards:** W
- **Move Backwards:** S
- **Turn Left:** A
- **Turn Right:** D

Interact:

The "E" key allows players to interact with various objects in the environment, such as talking to Zara, sitting on the park bench, collecting water from hidden sources, and discovering hidden items like ancient coins or artifacts.

Dialogue Choices:

During dialogue events with Zara, the player uses number keys (1, 2, 3) to select dialogue options. These choices can provide hints, directions, or background information that impacts how the player navigates the desert and progresses through the game.

Collect Water:

When near a water source, like a hidden water cache, players can press the "E" key to collect water and refill Amir's hydration meter, ensuring his survival in the harsh desert environment.

Survive Sandstorm:

During a sandstorm event, players must navigate Amir through low visibility using standard movement controls. Strategic movement is required to avoid hazards, find shelter, or reach the next checkpoint before Amir's health or hydration runs out.

Rules

1. Hydration Management:

Amir's hydration level gradually decreases over time due to the harsh desert environment. Players must find and collect water caches to refill the hydration meter. If the hydration level reaches zero, Amir will begin to lose health, and prolonged dehydration will trigger a respawn at the last checkpoint.

2. Health Monitoring:

Amir's health decreases when exposed to hazard, such as remaining too long in a sandstorm or running out of hydration. Players must avoid dangerous situations, find shelter, and manage resources to maintain health. If Amir's health reaches zero, the player will respawn at the last checkpoint with partially restored health and hydration.

3. Interaction Points:

The player must use the "E" key to interact with the NPC in the environment. Key interaction points include talking to Zara for hints, sitting at the park bench for rest, and collecting water caches.

4. Checkpoints and Respawn:

Checkpoint is located at a key point in the game, the park bench. If Amir's health or hydration reaches zero, he will respawn at the last checkpoint, with health and hydration partially restored.

5. Dialogue Choices:

During interactions with Zara or other dialogue events, players use number keys (1, 2, 3) to select responses. The choices made can influence the direction of the conversation, reveal new information, or provide hints and directions that impact gameplay.

6. Navigating Hazards:

Players must navigate through environmental hazards like sandstorms using standard movement controls. Failure to find shelter or a safe path in time may result in loss of health or a forced respawn. Hazards require quick thinking and strategic movement to survive.

7. Exploration and Rewards:

Exploring the desert environment is encouraged and rewarded. Players can find water caches, by thoroughly exploring. Discovering these items can increase the player's health.

8. Goal Achievement:

The primary objective is to guide Amir to the watchtower. Players must manage resources, interact with Zara for guidance, and overcome obstacles to reach the destination. Reaching the watchtower signifies the successful completion of the game.

World/Setting

"Desert Odyssey" is set in a wide, moonlit desert at night, with rolling sand dunes, rocky areas, and bits of dry vegetation scattered around. The desert feels both beautiful and tough, giving a sense of being alone in a mysterious place. Landmarks like an old park bench and a distant watchtower help guide the player and serve as key points to explore.

The environment changes as you play, with unexpected events like sandstorms that make it hard to see, forcing the player to think fast and move carefully. The world invites players to explore, find

hidden secrets, and survive the challenges, all while making their way to the final goal at the watchtower.



Figure 9: Desert World (Freepik)

Level Design

Scene Map

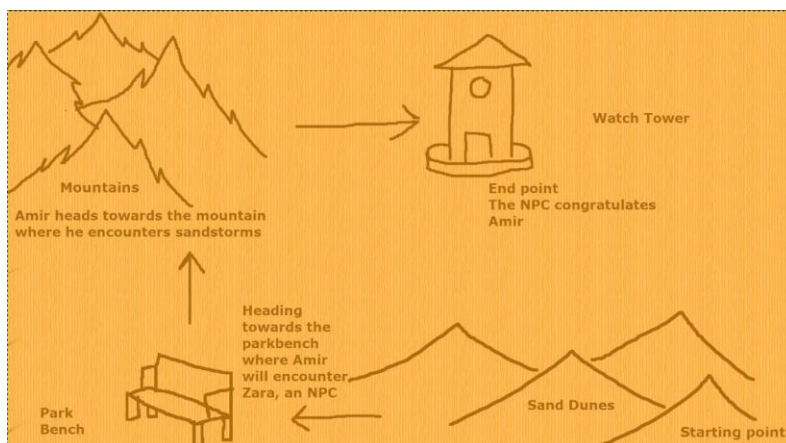


Figure 10: Scene Map (Own work, in progress)

Player Progression

- **Starting Point: Sand Dunes**

- The player begins at the edge of a vast desert, amidst endless sand dunes. The initial objective is to navigate these dunes while learning the basic controls and getting a feel for the environment.

- **Finding the First Water Cache**

- Early in the journey, Amir discovers the first water cache, a hidden resource to keep him hydrated. This introduces the player to the importance of resource management, ensuring Amir maintains hydration levels to avoid penalties or health loss.

- **Heading Towards the Park Bench**

- Guided by a faint outline in the distance, the player moves towards a park bench, the first key landmark.
- Reaching the bench, Amir meets Zara, an NPC who triggers a friendly dialogue, offering vital hints about the desert's challenges and advising on how to find additional water caches along the route.

- **Searching for More Water Caches**

- As Amir progresses towards the mountains, the player must remain vigilant for more water caches. These caches are scattered along the path, providing hydration and rewarding exploration.
- Players must decide whether to take direct routes or detour to search for these hidden caches to maintain hydration, adding strategic depth to the gameplay.

- **Navigating to the Mountains**

- With Zara's guidance, the player heads toward the mountains. As they approach, a sandstorm event is triggered, reducing visibility and creating an immediate challenge.
- The player must navigate carefully, using landmarks like the park bench, dunes, and distant watchtower, while managing hydration levels and the sandstorm's effects.

- **Facing the Sandstorm Challenge**
 - The sandstorm intensifies, requiring the player to make quick decisions and find temporary shelter or continue forward.
 - If Amir stays in the storm too long, he risks dehydration, making the discovery of water caches critical to his survival.
- **Reaching the Watchtower (End Point)**
 - After overcoming the sandstorm and finding necessary resources, the player reaches the watchtower.
 - Upon arrival, a final event is triggered where Zara congratulates Amir, marking the successful completion of his journey.
 - The watchtower offers a final vantage point and a moment of reflection on the challenging path taken.

Triggers/Events

- **NPC Interaction Dialogue**

Trigger: Player approaches and interacts with Zara, the NPC near the park bench.

Purpose: Provide hints, guidance, and information to help the player navigate through the desert challenges.

Pseudocode:

```
FUNCTION TRIGGER_NPC_INTERACTION
IF (player ENTERS interactionRange WITH Zara)
  DISPLAY "Dialogue UI with Zara's dialogue"
  GIVE "Hints and directions to player"
ENDIF
END FUNCTION
```

- **Sandstorm Event**

Trigger: Player reaches a specific location near the mountains.

Purpose: Create a challenging environment by reducing visibility, adding urgency, and testing the player's navigation skills.

Pseudocode:

```
VARIABLE sandstormActive = false
FUNCTION TRIGGER_SANDSTORM
IF (player ENTERS mountainArea)
    SET sandstormActive = true
    ACTIVATE "Sandstorm visual and audio effects"
    REDUCE "Player visibility"
    START sandstormTimer
ENDIF

WHILE (sandstormActive)
    IF (sandstormTimer > maxSandstormDuration)
        SET sandstormActive = false
        RESET "Visibility to normal"
    ENDIF
END WHILE
END FUNCTION
```

- **Player Respawn**

Trigger: Player's health depletes or they get lost due to the sandstorm or dehydration.

Purpose: Allow the player to respawn at the last checkpoint, maintaining game flow and offering another chance to progress.

Pseudocode:

```
FUNCTION TRIGGER_RESPAWN
IF (player health <= 0 OR player LOST)
    SET playerPosition = lastCheckpoint
    RESET "Player health and hydration"
    DISPLAY "Respawn message"
ENDIF
END FUNCTION
```

- **Water Cache Discovery**

Trigger: Player moves close to a hidden water cache location.

Purpose: Allow the player to collect water to maintain hydration, adding a strategic element to resource management.

Pseudocode:

```
VARIABLE waterCollected = false
FUNCTION TRIGGER_WATER_CACHE
```

```
IF (player ENTERS proximityToWaterCache)
  DISPLAY "Collect Water prompt"
  IF (player PRESSES "E" key)
    SET waterCollected = true
    UPDATE "Player hydration level"
    REMOVE "Water cache from scene"
  ENDIF
ENDIF
ENDIF
END FUNCTION
```

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